

STUDENT PERFORMANCE ANALYSIS

Using IBM SPSS Modeler

Dataset: Student Performance Dataset

Submitted By:

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Course: BCA (Data Science & Artificial Intelligence)

Semester: V

Software Used: IBM SPSS Modeler

Introduction:

This project aims to predict student academic performance using IBM SPSS Modeler. The student dataset is analyzed using data preparation, partitioning and a C&R Tree classification/modelling technique. The target field used in the project is Performance, which classifies students into Pass and Fail categories.

1. Objective: To analyze student performance using IBM SPSS Modeler and classify students into Pass and Fail categories using a decision tree model.

2. Dataset: Student Performance Dataset

Total Records: 395

Target Variable: Performance

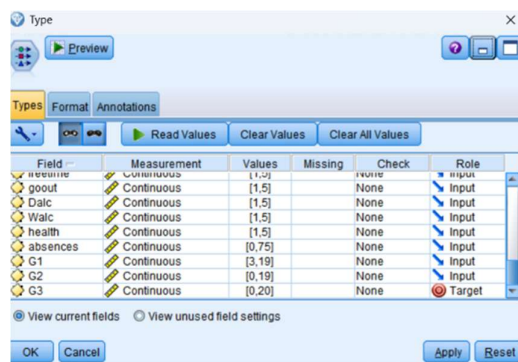
Target Classes: Pass, Fail

3. Tools Used

IBM SPSS Modeler, C&R Tree (Decision Tree), Student Performance Dataset

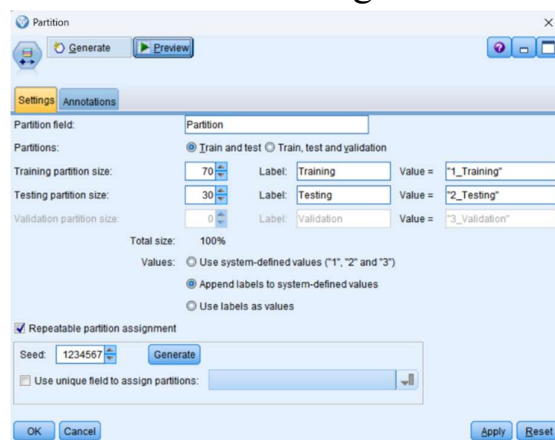
4. Data Preparation

The student dataset was imported into IBM SPSS Modeler from a CSV file. G3 was selected as the target field for the C&R Tree model.



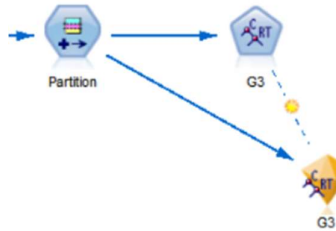
5. Data Partitioning

The dataset was divided into training and testing data using the Partition node. 70% for training and 30% for testing.



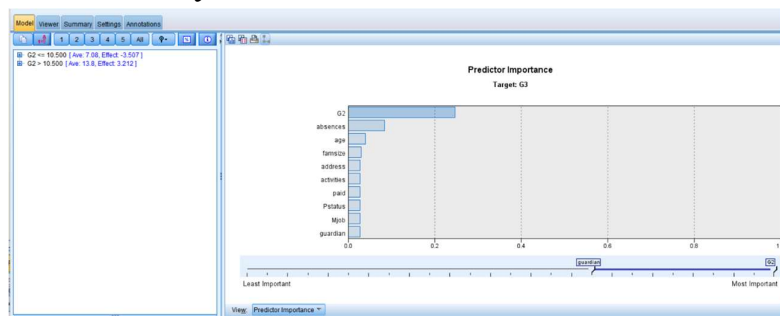
6. C&R Tree Model

The C&R Tree model was used to build a prediction model for the target field G3. The model was trained using the training data and evaluated using the testing data.



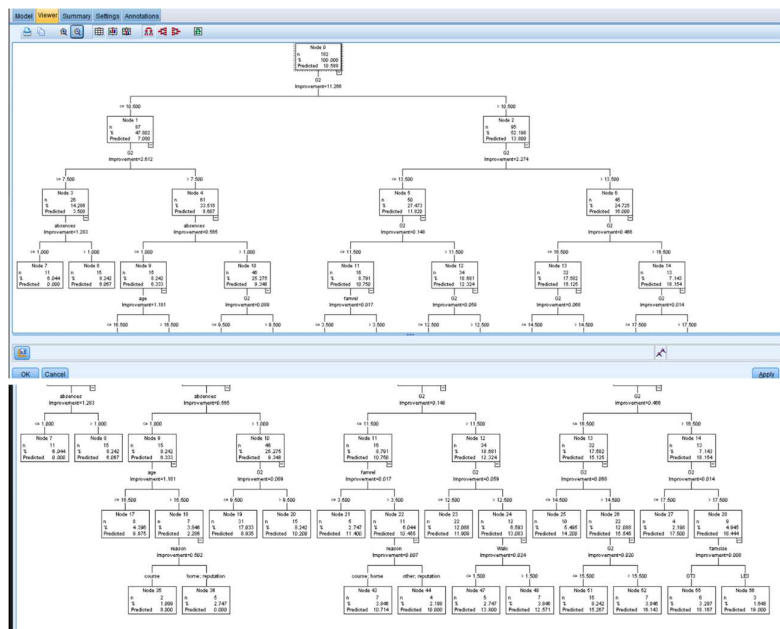
7. Model Output- Diamond

After running the C&R Tree model, a model nugget (diamond) was generated. It contains the trained model and provides the model output for further analysis.



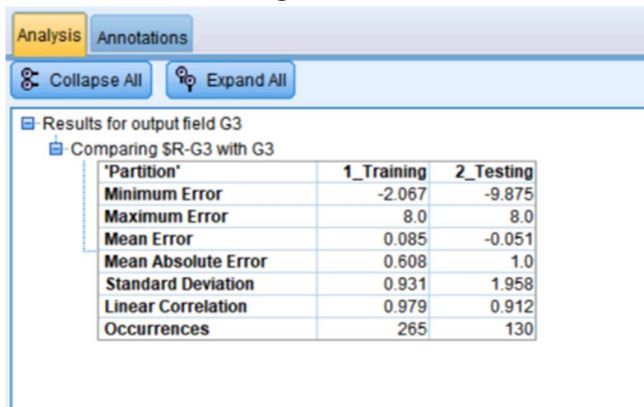
8. Tree Structure / Viewer

The Viewer displays the tree structure generated by the C&R Tree model.



9. G3 Analysis

The Analysis node was used to evaluate the predictions of the C&R Tree model for the target field G3.



Analysis Annotations

Collapse All Expand All

Results for output field G3

Comparing \$R-G3 with G3

'Partition'	1_Training	2_Testing
Minimum Error	-2.067	-9.875
Maximum Error	8.0	8.0
Mean Error	0.085	-0.051
Mean Absolute Error	0.608	1.0
Standard Deviation	0.931	1.958
Linear Correlation	0.979	0.912
Occurrences	265	130